

- 9-4** Find the Laplace transform of $f(t) = A[\cos(\beta t) - \sin(\beta t)]u(t)$. Locate the poles and zeros of $F(s)$.
- 9-5** Find the Laplace transform of $f(t) = 2\delta(t) - 100e^{-10t} \cos(t)u(10t)$. Locate the poles and zeros of $F(s)$.
- 9-6** Find the Laplace transform of $f(t) = 30[2 - 5t - 2e^{-5t}]u(t)$. Locate the poles and zeros of $F(s)$.
- 9-7** Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams:
- (a) $f_1(t) = [15e^{-5t} - 20e^{-10t}]u(t)$
 - (b) $f_2(t) = [5 \cos 10t + 10 \cos 20t]u(t)$
- 9-8** Find the Laplace transforms of the following waveforms and plot their pole-zero diagrams.
- (a) $f_1(t) = 4\delta(t) + [20e^{-20t} + 40e^{-40t}]u(t)$
 - (b) $f_2(t) = [15 + 15 \cos 500t]u(t)$
-  **9-9** Use MATLAB to find the Laplace transforms of the following waveforms. Locate the poles and zeros of $F(s)$.

9-16 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{s + 100}{s(s + 50)}$

(b) $F_2(s) = \frac{s + 10}{s(s + 50)(s + 100)}$

9-17 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{50(s + 10)}{(s + 5)(s + 20)}$

(b) $F_2(s) = \frac{2s^2}{(s + 200)(s + 500)}$

9-18 Find the inverse Laplace transforms of the following functions:

9-19 Find the inverse Laplace transforms of the following functions and sketch their waveforms for $\beta > 0$:

(a) $F_1(s) = \frac{\beta(s + \beta)}{s(s^2 + \beta^2)}$

(b) $F_2(s) = \frac{s(s + \beta)}{s^2 + \beta^2}$

9-20 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{\alpha^2}{s^2(s + \alpha)}$

(b) $F_2(s) = \frac{\alpha^2}{s(s + \alpha)^2}$

9-21 Find the inverse Laplace transforms of the following functions:

(a) $F_1(s) = \frac{600}{(s + 10)(s + 20)(s + 40)}$

(b) $F_2(s) = \frac{6(s + 10)}{(s + 5)(s + 20)}$

10-1 Find $Z_{EQ}(s)$ in Figure P10-1. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

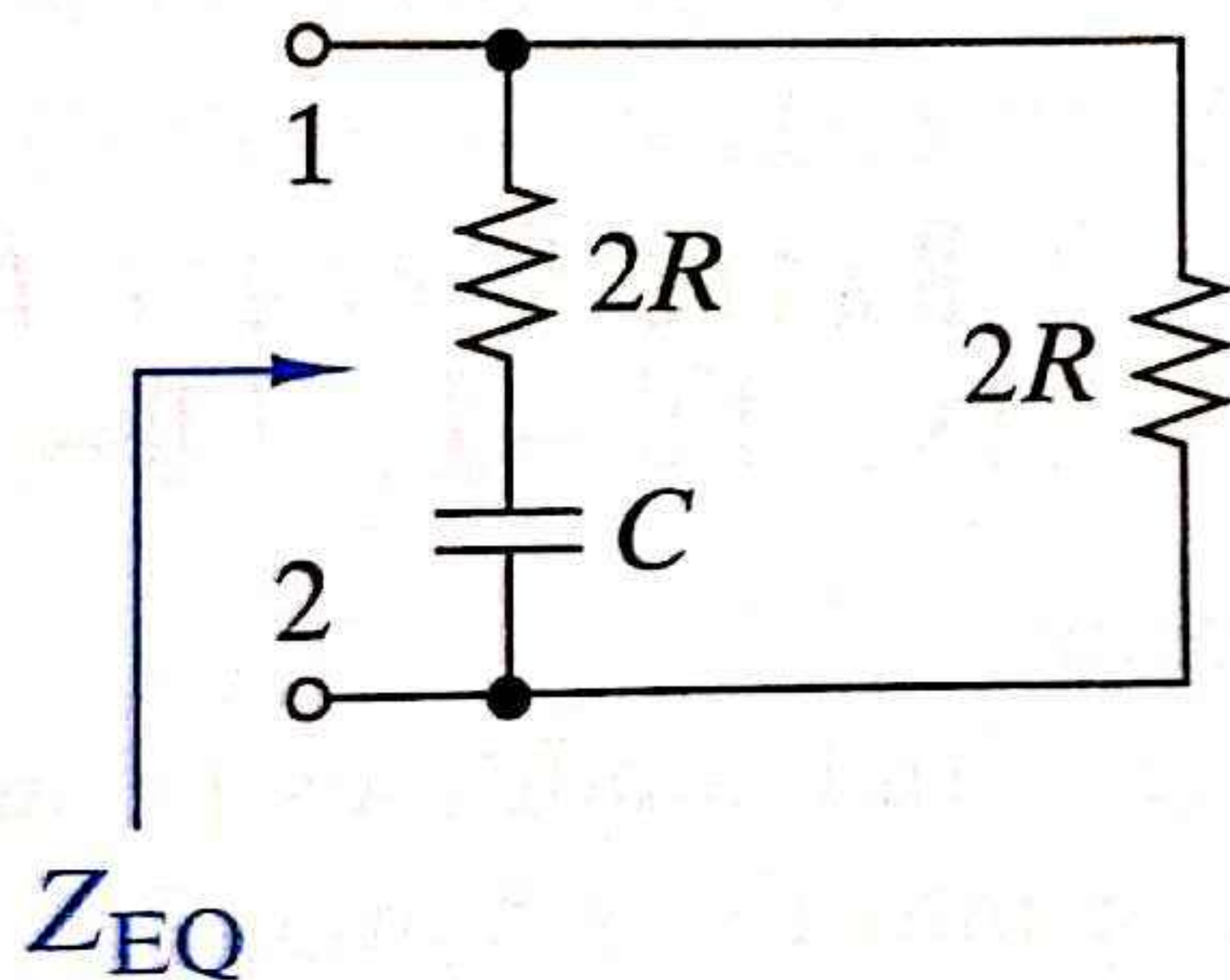


FIGURE P10-1

10-2 Find $Z_{EQ}(s)$ in Figure P10-2. Express $Z_{EQ}(s)$ as a rational function and locate its poles and zeros.

10-3 Find $Z_{EQ}(s)$ as a rational function and locate its poles and zeros. $L = 0.4 \text{ H}$, $C = 0.1 \text{ F}$.

Z_{EQ1} Z_{EQ2}

Figure P10-6 with

FIGURE P10-9

- 10-10 **D** Find the equivalent impedance between terminals 1 and 2 in Figure P10-10. Select values of R and L so that $Z_{EQ}(s)$ has a pole at $s = -2000$ rad/s. Locate the zeros of $Z_{EQ}(s)$ for your choice of R and L .

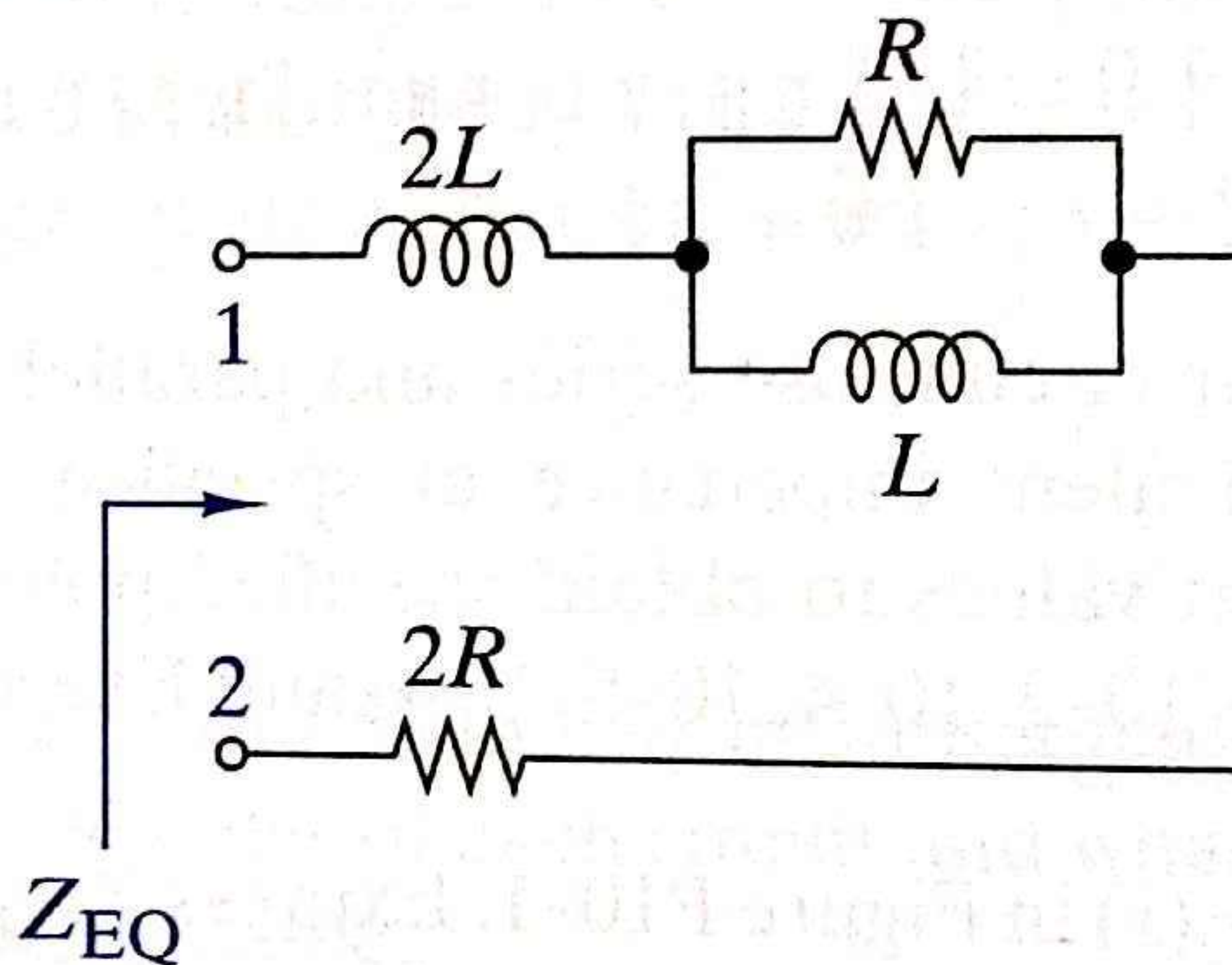


FIGURE P10-10

7. Express each
its poles and

10-14 The switch in Figure P10-13 has been in position B for a long time and is moved to position A at $t = 0$. Transform the circuit into the s domain and solve for $V_C(s)$ and $v_C(t)$ in symbolic form.

10-15 Transform the circuit in Figure P10-15 into the s domain and find $I_L(s)$, $i_L(t)$, $V_L(s)$, and $v_L(t)$ when $v_1(t) = V_A u(t)$ and $i_L(0) = I_A$.

10-20 T
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put V

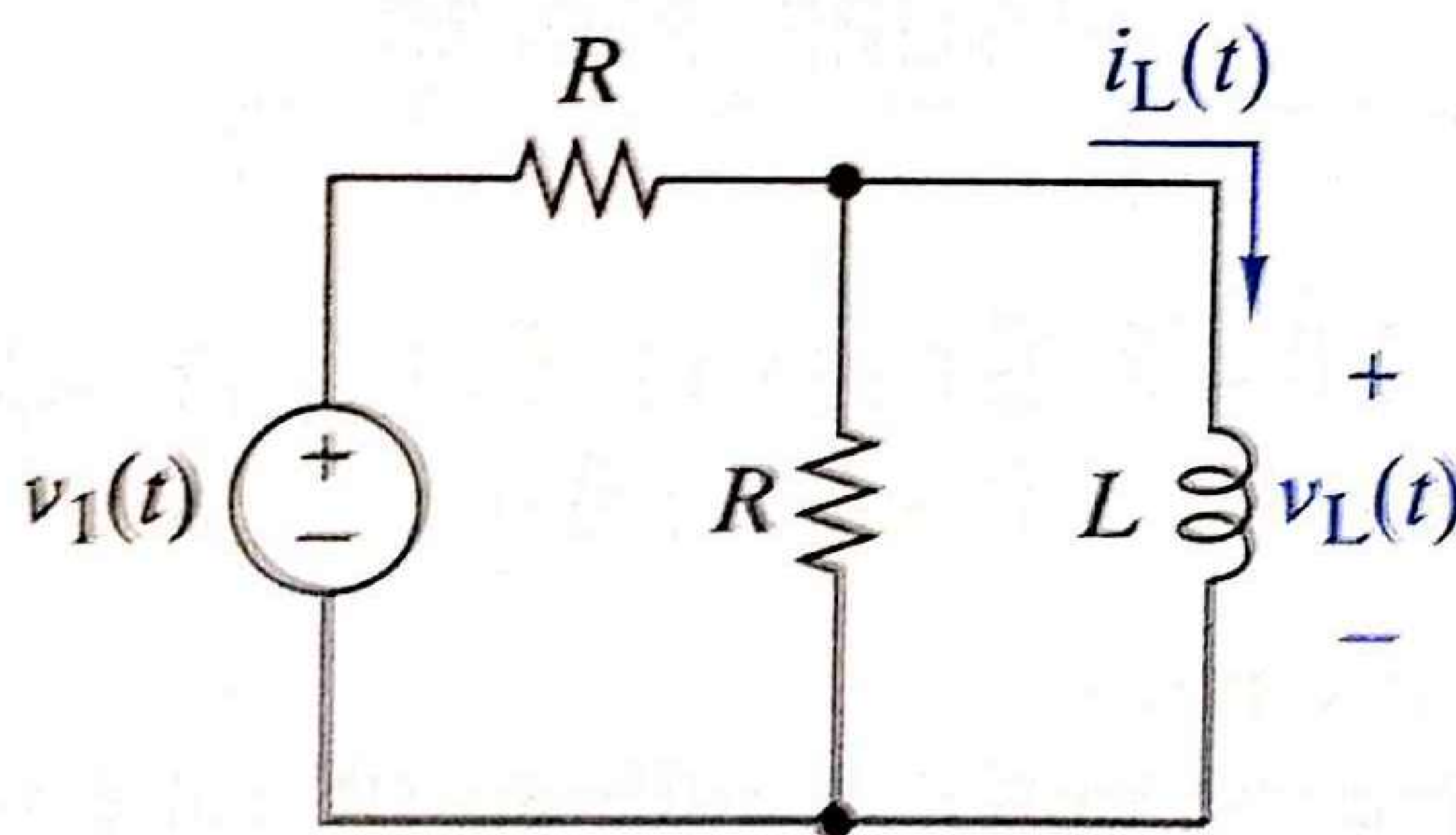


FIGURE P10-15

param-
s,
10-8,
10-7,

10-16 Transform the circuit in Figure P10-15 into the s domain and find $I_L(s)$, $i_L(t)$, $V_L(s)$, and $v_L(t)$ when $v_1(t) = V_A e^{-\omega t} u(t)$ and $i_L(0) = I_A$.

10-17 The switch in Figure P10-17 has been in position A for a long time and is moved to position B at $t = 0$.

(a) Transform the circuit into the s domain and solve for $I_L(s)$ in symbolic form.

(b) Repeat part (a) using MATLAB.

(c) Find $i_L(t)$ for $R_1 = R_2 = 500 \Omega$, $R_3 = 1 \text{ k}\Omega$, $L = 250 \text{ mH}$, $C = 4 \mu\text{F}$, and $V_A = 15 \text{ V}$.

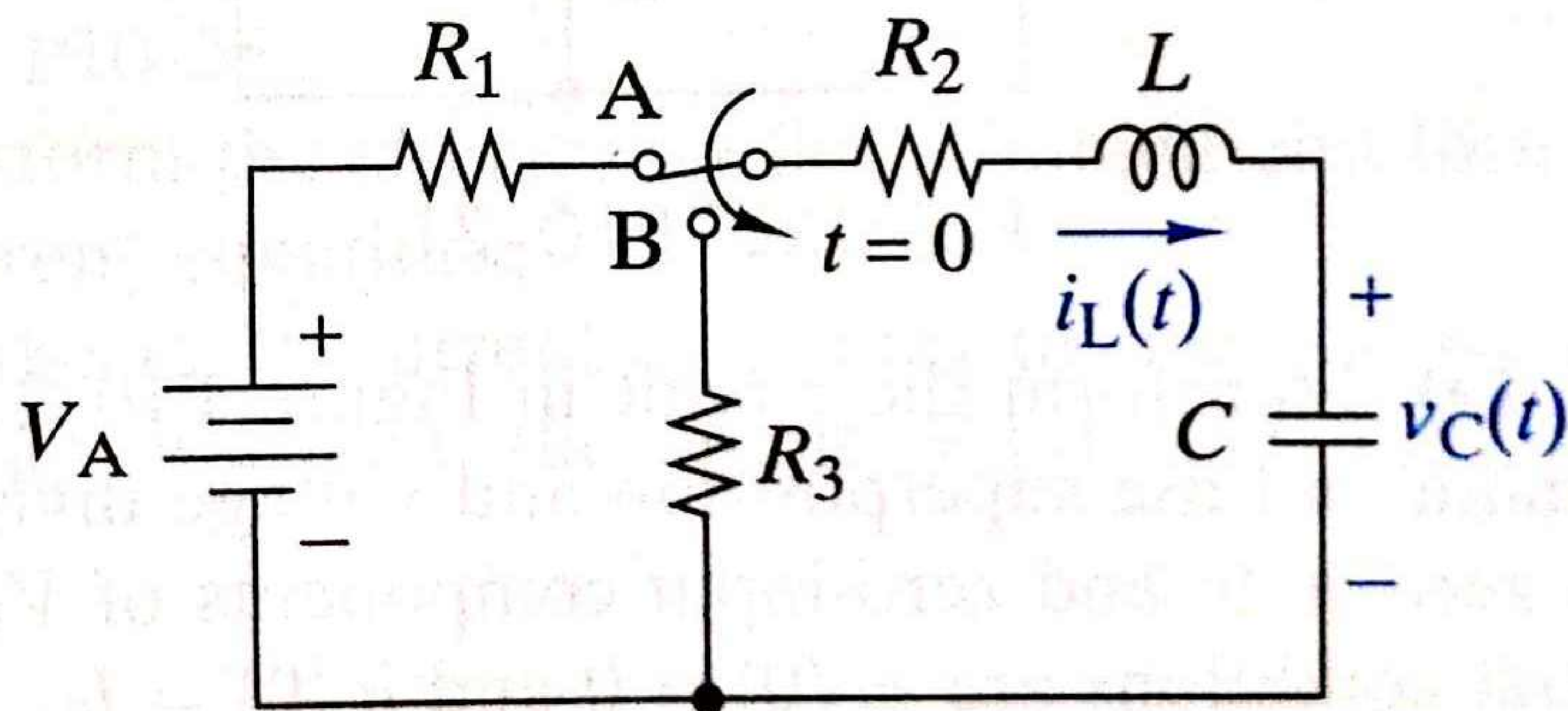


FIGURE P10-17

B for a
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10-18 The switch in Figure P10-17 has been in position B for a

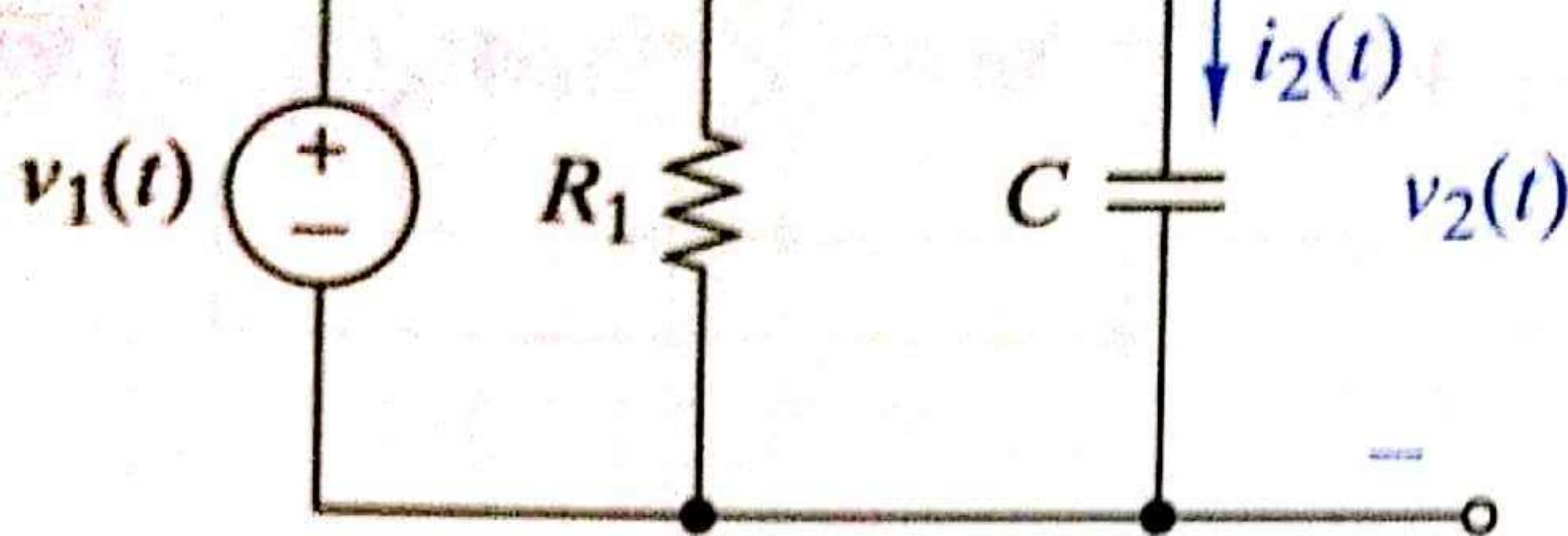


FIGURE P10-31

10-32 There is no initial energy stored in the circuit in Figure P10-31.

(a) Transform the circuit into the s domain and formulate node-voltage equations.

(b) Solve these equations for $V_2(s)$ in symbolic form.

(c) Identify the natural and forced poles of $V_2(s)$.

(d) Find $v_2(t)$ for $v_1(t) = 15u(t)$ V, $R_1 = R_2 = 500 \Omega$, $L = 0.5$ H, and $C = 2 \mu\text{F}$.

10-33 There is no initial energy stored in the circuit in Figure P10-33.

(a) Transform the circuit into the s domain and formulate node-voltage equations.

(b) Solve these equations for $V_2(s)$ in symbolic form.

(c) Insert a buffer at point A and solve for $V_2(s)$ in symbolic form. How does the buffer change the denominator

10-36 T

in Fig

(a) T

node

(b) U

relati

 **10-37** T

P10-3

(a) Fi

$v_1(t) =$

(b) V